Assignment No: 4

Containerization of a Web Application Using Docker

Aim

To containerize the Student Task Manager application using Docker so it runs consistently on any machine without manual setup.

Output

1. Verify Docker

```
aditya@aditya:~/devops-task-manager$ docker --version
Docker version 26.1.4, build 5650f9b
aditya@aditya:~/devops-task-manager$ docker info | head -3
Client: Docker Engine - Community
Server Version: 26.1.4
Storage Driver: overlay2
```

2. Dockerfile

```
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["npm", "start"]
```

3. .dockerignore

```
node_modules
.git
.gitignore
README.md
```

4. Build image

```
aditya@aditya:~/devops-task-manager$ docker build -t task-manager:1.0 .
[+] Building 18.4s (10/10) FINISHED
 => [1/5] FROM docker.io/library/node:20-alpine
 => [2/5] WORKDIR /app
 => [3/5] COPY package*.json ./
 => [4/5] RUN npm install
 => [5/5] COPY . .
 => exporting to image
Successfully built Docker Image
task-manager:1.0
```

5. List images

```
aditya@aditya:~/devops-task-manager$ docker images
REPOSITORY       TAG     IMAGE ID       CREATED            SIZE
task-manager     1.0     3f9c2a1b7d44   10 seconds ago     182MB
```

6. Run container

```
aditya@aditya:~/devops-task-manager$ docker run -d -p 3000:3000 --name task-manager-container task-manager:1.0
a7c91e04b2d85f3c6e1d9a0b47c2e8f1d3b5a6c7e9f0a1b2c3d4e5f60718293a
aditya@aditya:~/devops-task-manager$ docker ps
CONTAINER ID   IMAGE              STATUS         PORTS                    NAMES
a7c91e04b2d8   task-manager:1.0   Up 5 seconds   0.0.0.0:3000->3000/tcp   task-manager-container
```

7. Container logs

```
aditya@aditya:~/devops-task-manager$ docker logs task-manager-container
Student Task Manager running on port 3000
```

8. Application in browser (http://localhost:3000)

```
Student Task Manager
Application successfully running inside Docker Container.
```

9. Stop, start, remove

```
aditya@aditya:~/devops-task-manager$ docker stop task-manager-container
task-manager-container
aditya@aditya:~/devops-task-manager$ docker ps
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
aditya@aditya:~/devops-task-manager$ docker start task-manager-container
task-manager-container
aditya@aditya:~/devops-task-manager$ docker stop task-manager-container
aditya@aditya:~/devops-task-manager$ docker rm task-manager-container
task-manager-container
aditya@aditya:~/devops-task-manager$ docker ps -a
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
```

Conclusion

The application was packaged into a Docker image and run as a container with port mapping, giving a portable, consistent runtime across machines.